

Emil Manninen

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Self-taught developer with broad fullstack capability (React/Next.js, Python/FastAPI, PostgreSQL, Docker), actively seeking a junior developer role. Builds production-shaped projects end to end, including RAG/AI pipelines, authentication systems, and CI/CD. Proficient with AI-assisted development tools (e.g. Claude) as part of a modern engineering workflow, while still manually reviewing and understanding shipped lines. Background in high-level competitive esports underpins a disciplined, iterative approach to learning new skills quickly.

PROJECTS

RAG Document Q&A Bot *FastAPI, PostgreSQL/pgvector, Voyage AI, Anthropic API, Next.js, shadcn/ui*

Hand-rolled (no LangChain) retrieval-augmented Q&A chatbot over Finnish government (Kela) student-aid documentation, with per-answer citations.

- Built a swappable chunking-strategy layer and provider-abstracted LLM/embedding clients; migrating LLM providers and adding query condensation for multi-turn chat both required zero changes to the retrieval layer, by design.
- Designed and ran a retrieval eval harness (Recall@k) across 3 chunking strategies on a 38-question hand-built test set, tracing every metric gap to its root cause (misabeled ground truth, ambiguous phrasing, or a genuine chunking difference) rather than reporting raw numbers.
- Built query condensation for multi-turn conversations, tested against adversarial cases (vague follow-ups, ambiguous references) before wiring into the app, so retrieval doesn't drift on follow-up questions like “what about for parents?”

Live: qaragbot.vercel.app Code: github.com/emilmanninen/qaragbot

League of Legends Analytics Pipeline *Python, PostgreSQL, Next.js, shadcn/ui*

End-to-end data pipeline: a Python crawler ingests match data into PostgreSQL, surfaced through a separate Next.js dashboard.

- Built a crawler that gathered a substantial match dataset, with a deliberate ingestion/storage/presentation split across two repos rather than one monolith.

Live: leaguefrontend.vercel.app Frontend: github.com/emilmanninen/leaguefrontend Pipeline: github.com/emilmanninen/leaguepipeline

ToolShare *Next.js, TypeScript, Supabase (PostgreSQL), Vitest, GitHub Actions*

Full-stack neighborhood tool-lending marketplace, deployed live on Vercel.

- Used Postgres Row-Level Security as the primary authorization layer rather than app-level checks, reducing the surface area for access-control bugs.
- Designed the data model to reuse a listing_type column across tool and item listings instead of separate tables, avoiding schema duplication as the app grew.
- Automated testing (Vitest) and CI/CD (GitHub Actions) from early in the project, not bolted on after.

Live: neighborhoodapp-xi.vercel.app Code: github.com/emilmanninen/neighborhoodapp

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, SQL

Frontend: React, Next.js, shadcn/ui, Vite

Backend: Node.js/Express, FastAPI, REST API design, WebSockets

Data & Infra: PostgreSQL, pgvector, Supabase, SQLite, Docker, GitHub Actions (CI/CD), AWS (EC2, S3, ECS, RDS, Lambda, IAM)

AI/RAG: Retrieval-augmented generation, vector search, embeddings, Anthropic/Gemini APIs

Other: Git, automated testing (Vitest), Linux/WSL

EXPERIENCE

Restaurant Worker, Hesburger 2023 – 2025

Customer service and day-to-day restaurant operations in a fast-paced environment.

Competitive Esports — Marvel Rivals 1.5 years

Reached world-class ranked standing; independently formed and captained team “dubaimatcha,” placing top 16 at MRC Season 4: Heart of the Dragon (EMEA, PC). Built a disciplined daily practice routine around performance review and iterative improvement — the same self-directed process now applied to software development.

EDUCATION & COURSEWORK

Data Structures & Algorithms *Boot.dev*

Completed a project-based DSA course — implementation-level work, not video-only learning.

Cloud Computing (AWS) *Boot.dev*

Project-based course covering core AWS services — EC2, S3, IAM, ECS, RDS, and serverless (Lambda).

Information Systems Sciences (partial studies) *University of Jyväskylä*

Coursework covering foundational programming and how systems work behind applications, websites, and servers.

LANGUAGES

Finnish (native), English (fluent), Swedish (basic), Spanish (basic)